

Nyx Haile

Founder & Systems Engineer

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EXPERIENCE

Partial

2025 – Present

Cofounder & CEO

London

- Designed a perpetual prediction-market exchange where positions are probability distributions over joint price paths across correlated assets, not binary outcomes.
- Specified eleven market types, among them price paths, derived statistics (correlations, lead-lag) and reported statistics, under a single mechanism-design framework.

Vana

Dec 2022 – July 2023

AI Developer

Remote

- Engineered and iterated prompt templates for personalised AI-generated sticker portraits, tuning style and expression cues for identity retention and consistency.

SELECTED PROJECTS

Repositories marked private are available on request.

Strand (private) | Rust, FoundationDB

Capability-authenticated issue tracker

- Built a federable, capability-authenticated issue tracker in Rust on FoundationDB, with line-level blame and content-addressed rationale.
- Shipped a 16-crate, ~71k-line Rust workspace through v1.0.0 GA (10/10 verification gates) and v1.1.0, then a v2.0 governed-execution prerelease.

Forge (private) | Rust, systemd, RPC

Local-first agent control plane

- Built a Rust daemon, operator CLI, Claude Code hooks and provider proxy that coordinate coding-agent work across multiple git repositories.
- Added bounded, jittered exponential backoff to scheduler dispatch, seeded per task so retry waves de-correlate in production and replay exactly in tests.

Astraia (private) | TypeScript, Python, SQLite

Personal multi-agent operations system (on OpenClaw)

- Built a localhost-only operations dashboard (Bun, React, SQLite) where outbound email and messages pass expiring, hash-pinned previews and fail closed.

howtwin.lol | Python, PyTorch, CUDA, Redis

World model for post-game teaching

- Designed a Dreamer-lineage RSSM world model for League of Legends combining static context, player priors and event sequences with outcome, next-event, next-frame and next-decision heads.

MIT Pokerbots | Python

Heads-up poker-variant bot, January 2026

- Built a Python bot for MIT Pokerbots' heads-up poker variant, largely with AI-assisted coding; the team placed about 15th of 70.
- Implemented Monte Carlo equity estimation, with exact enumeration for discards, using quick and deep sampling paths to fit a per-decision compute budget.

Sherlock | Rust, Dolt

Token-usage forensics for agent CLIs

- Built a Rust CLI that ingests agent-CLI logs (Claude Code, Codex, Copilot), attributes token usage per session, and reports what burns tokens without adding value.

Conch | Rust, ONNX Runtime

Voice-interview-to-brief CLI

- Built a Rust voice-interview CLI that runs spoken Q&A over mic and TTS at three model depths and writes a persisted brief.

EDUCATION

Massachusetts Institute of Technology

2026

B.S. Computer Science, B.S. Mathematics

Cambridge, MA

- Coursework: algorithms, theory of computation, computer systems security, cryptography, statistics.

TECHNICAL SKILLS

Languages: Rust, Python, TypeScript, SQL **ML:** PyTorch, NumPy, CUDA, GPU training and retrieval **Systems:** Linux/systemd, Docker, FoundationDB, PostgreSQL, SQLite, Redis **Web:** SvelteKit, React, Next.js